# Design Calculation Sheet - VBF-T3R2-CALC-01 (PDF release of calc.xlsx)

| Parameter | Value | Unit | Source |
|---|---|---|---|
| Sheet: Inputs | | | |
| Positive electrode thickness [m] | 4.5e-05 | m | design override (r2_p1_params_final.json) |
| Negative electrode thickness [m] | 5.6e-05 | m | design override |
| Positive electrode porosity | 0.45 | - | design override |
| Negative electrode porosity | 0.4 | - | design override |
| Separator thickness [m] | 1.2e-05 | m | Chen2020 parameter set |
| Separator porosity | 0.47 | - | Chen2020 parameter set |
| Positive particle radius [m] | 1.3e-06 | m | design override (5.22 um base) |
| Negative particle radius [m] | 1.6e-06 | m | design override (5.86 um base) |
| Electrolyte conductivity [S.m-1] | 1.5 | S/m | design override, constant (replaces Nyman2008 function) |
| Electrolyte diffusivity [m2.s-1] | 6e-10 | m2/s | design override, constant |
| Nominal cell capacity [A.h] | 3.3219 | Ah | re-based to delivered 1C capacity (audited rule) |
| Electrode height x width [m] | 0.065 x 1.58 | m | Chen2020 parameter set |
| Active material volume fraction pos/neg | 0.665 / 0.75 | - | Chen2020 parameter set (model solid fraction) |
| Upper / lower voltage cut-off [V] | 4.2 / 2.5 | V | Chen2020 parameter set |
| Ambient temperature 1C/5C discharge | 298.15 | K | protocol |
| Ambient temperature 4C charge | 318.15 | K | protocol (45 C) |
| Sheet: Capacity & energy | | | |
| 1C discharge capacity | 3.3278 | Ah | cell/r2_p1_1c_dfn.json:capacity_ah |
| 5C discharge capacity | 3.2911 | Ah | cell/r2_p1_5c_dfn.json:capacity_ah |
| 5C capacity retention | 0.9889 | - | 5C cap / 1C cap (cell/r2_p1_derived.json) |
| Rated energy (1C integral of V*I) | 12.1255 | Wh | cell/r2_p1_energy.json:energy_wh |
| Discharge midpoint voltage | 3.8855 | V | cell/r2_p1_energy.json:midpoint_voltage_v |
| DCR (V_start - V@10%) / I_1C | 2.2654 | mOhm | cell/r2_p1_energy.json:dcr_ohm |
| Power density V_OC^2/(4 DCR)/mass | 63129.7 | W/kg | cell/r2_p1_energy.json:power_density_w_kg |
| 4C charge T_max | 332.5213 | K | cell/r2_p1_4c_dfn.json:T_max_K |
| 4C anode potential min | 0.03641 | V | cell/r2_p1_4c_dfn.json:anode_potential_v (>= 0 -> no plating) |
| 5C discharge T_max | 322.2243 | K | cell/r2_p1_derived.json:t_max_5c_k |
| Sheet: Energy density | | | |
| Layer masses (pos / neg / AlCC / CuCC / sep) | 8.29 / 5.72 / 4.44 / 11.04 / 0.26 | g | calc-energy layer_kg_m2 x area x 1000 |
| Total mass (contract caliber, electrolyte excluded) | 29.75 | g | cell/r2_p1_energy.json:mass_kg |

| Parameter | Value | Unit | Source |
|---|---|---|---|
| Energy density (Wh/kg) | 407.6 | Wh/kg | energy_wh / mass_kg |
| Volumetric energy density (Wh/L) | 837.4 | Wh/L | energy_wh / (total layer thickness 141 um x area) |
| Total layer stack thickness | 141 | um | pos 45 + sep 12 + neg 56 + Al 16 + Cu 12 |
| Electrolyte mass (informative, lit. density 1.2 g/cm3) | 5.95 | g | pore volume x density; excluded from ED caliber |
| Sheet: N/P & mass | | | |
| N/P theoretical full-range (neg cap.density x thickness / pos ...) | 0.74 | - | 33133 x 0.75 x 56e-6 / (63104 x 0.665 x 45e-6), mol/m3 formula caliber |
| Positive full theoretical areal capacity | 50.62 | Ah/m2 | 63104 x F x 1 x 45e-6 x 0.665 / 3600 |
| Negative full theoretical areal capacity | 37.29 | Ah/m2 | 33133 x F x 1 x 56e-6 x 0.75 / 3600 |
| Pos stoichiometry start -> end (1C discharge) | 0.2700 -> 0.9102 | - | _logs/p1_stoich_probe.json |
| Neg stoichiometry start -> end (1C discharge) | 0.9014 -> 0.0326 | - | _logs/p1_stoich_probe.json |
| Consumed areal capacity pos = neg = cell | 32.403 | Ah/m2 | probe identity check 1.0000 / 1.0000 (series current balance) |
| Anode lithiation headroom at full charge | 9.86 % | - | (1 - 0.9014) from probe |
| Mass per energy (contract caliber) | 2.453 | kg/kWh | total g / energy_wh |
| Sheet: Process parameters | | | |
| Positive areal density | 80.73 | g/m2 | thickness x (1 - porosity) x density 3262 |
| Negative areal density | 55.68 | g/m2 | thickness x (1 - porosity) x density 1657 |
| Positive compaction density | 1.794 | g/cm3 | 3262 x (1 - 0.45) / 1000 |
| Negative compaction density | 0.994 | g/cm3 | 1657 x (1 - 0.40) / 1000 |
| Electrolyte fill amount | 5.95 | g | pore volume 4.96 cm3 x 1.2 g/cm3 (literature density) |
| Formation recommendation | 0.1C CC to 4.2 V, 25 C, 2 cycles | - | design recommended value; production tuning required |
| Nominal capacity re-basing note | nominal := measured 1C capacity (iterated, drift < 2%) | - | audited design rule, log plan_update_r1 |